UNIT

ORDINARY DIFFERENTIAL EQUATIONS

PAGE NO.
DATE / /

- Differential equation is an equation which involves differentials or differential coefficients.

* ORDINARY DIFFERENTIAL EQUATION (O.D.E.)

- Differential equation which involves only ordinary derivative — O.D.E.

Eg) $\frac{d^2y}{dx^2} - 2\frac{dy}{dx} - 3y = 5x$

- Order of D.E. is the order of highest derivative involved in D.E.
- Degree of D.E. is degree of highest derivative involved in D.E., where all derivative occuring therein are free from radicals and fraction.

Eg) $\frac{d^3y}{dx^3} + 3\frac{d^2y}{dx^2} + \frac{dy}{dx} + 3 = \cos x$ ⇒ Order : 3, Degree : 1

- $\frac{\left[1+\left(\frac{dy}{dx}\right)^2\right]^{3/2}}{\frac{d^2y}{dx^2}} = C$

$\Rightarrow \left(1+(y')^2\right)^{3/2} = Cy''$

Sq. both sides $\Rightarrow \left(1+(y')^2\right)^3 = C^2(y'')^2$

∴ Order = 2
degree : 3 //

- $\sqrt[4]{(y'')^5} = \sqrt{7+3(y')^2}$

$(y'')^{5/4} = \left(7+3(y')^2\right)^{1/2}$

$(y'')^5 = \left(7+3(y')^2\right)^2$

Order = 2
Degree = 5

Scanned with OKEN Scanner

* Solution (Integral or Primitive) of D.E.

It is the relation btw independent & dependent variable which does not involve any derivatives and also satisfies given D.E.

Eg) $y = e^x$ is a solution of $y'' - 2y' + y = 0$.

* Types of Solution :

i) A solution of D.E. in which no. of arbitary const. is equal to order of D.E. → General solution or complete integral.

ii) Solution obtained from general solution by given particular values to arbitary constant is called particular solution.

iii) A solution which cannot be obtained from general solution → singular solution.

Methods to find Solution of D.E. 1st order & 1st degree

i) Variable Seperable method :

General eqn : $N(y)\,dy = M(x)\,dx$

Soln : $\int N(y)\,dy = \int M(x)\,dx + C$

Eg) $\frac{dy}{dx} = \sec^2 y$

$\Rightarrow \frac{dy}{\sec^2 y} = dx$

$\cos^2 y\,dy = dx \Rightarrow \int \frac{1+\cos 2y}{2}\,dy = \int dx$

Scanned with OKEN Scanner

$$\therefore \frac{y}{2} + \frac{\sin(2y)}{4} = x + c$$

$$\Rightarrow y + \frac{\sin(2y)}{2} = 2(x+c)$$

Eg: $e^x \frac{dy}{dx} = e^{2x-2y} + x^2 e^{-2y}$

EXACT DIFFERENTIAL EQUATION

- A D.E. of the form $Mdx + Ndy = 0$ is said to be exact if it is the exact differential of some function 'u' i.e., $\boxed{du = Mdx + Ndy}$.

- Condition for exactness:

* A D.E $Mdx + Ndy = 0$ is said to be exact if,

$$\boxed{\frac{\partial M}{\partial y} = \frac{\partial N}{\partial x}}$$

- Solution of an exact D.E. is given by,

$$\int M\,dx + \int (\text{terms of } N \text{ not containing } x)\,dy = C$$

Scanned with OKEN Scanner

Q) Solve:

i) $(3x^2+6xy^2)\,dx + (6x^2y+4y^3)\,dy = 0$

→ Given:

$(3x^2+6xy^2)\,dx + (6x^2y+4y^3)\,dy = 0$

This is in the form $Mdx + Ndy = 0$ where

$M = 3x^2+6xy^2$

$N = 6x^2y+4y^3$

$\frac{\partial M}{\partial y} = 12xy \quad ; \quad \frac{\partial N}{\partial x} = 12xy$

$\Rightarrow \boxed{\frac{\partial M}{\partial y} = \frac{\partial N}{\partial x}}$ given D.E. is exact

∴ Solution is

$\int M\,dx + \int (\text{Terms of } N \text{ not containing } x)\,dy = c$

$\int (3x^2+6xy^2)\,dx + \int 4y^3\,dy = c$

$(x^3+3x^2y^2) + y^4 = c$

$\Rightarrow \boxed{x^3+3x^2y^2+y^4 = c}$

ii) $\left(y\left(1+\frac{1}{x}\right)+\cos y\right)dx + (x+\log x - x\sin y)\,dy = 0$

→ Given:

$\left(y\left(1+\frac{1}{x}\right)+\cos y\right)dx + (x+\log x - x\sin y)\,dy = 0$

This is in the form $Mdx + Ndy = 0$ where

$M = y\left(1+\frac{1}{x}\right)+\cos y$

$N = x+\log x - x\sin y$

Scanned with OKEN Scanner

PAGE NO.
DATE / /

$\frac{\partial M}{\partial y} = 1 + \frac{1}{x} - \sin y$ $\quad$ $\frac{\partial N}{\partial x} = 1 + \frac{1}{x} - \sin y$

$\frac{\partial M}{\partial y} = \frac{\partial N}{\partial x}$ function is exact.

The solution,

$\int M\,dx + \int$ (Terms of N not containing x) $dy = C$

$\int \left(y + \frac{y}{x}\right) dx + \int 0\,dy = C$

$\Rightarrow$ ~~$yx + y\log x = C$~~ $\quad$ $yx + y\log x + \cos y = C$

is the required solution.

iii) $\frac{dy}{dx} + \frac{y\cos x + \sin y + y}{\sin x + x\cos y + x} = 0$

$\rightarrow \frac{dy}{dx} = \frac{-y\cos x - \sin y - y}{\sin x + x\cos y + x}$

$\Rightarrow (\sin x + x\cos y + x)\,dy = (-y\cos x - \sin y - y)\,dx$

$\Rightarrow (y\cos x + \sin y + y)\,dx + (\sin x + x\cos y + x)\,dy = 0$

$M = y\cos x + \sin y + y$ $\quad$ $N = \sin x + x\cos y + x$

$\frac{\partial M}{\partial y} = \cos x + \cos y + 1$ $\quad$ $\frac{\partial N}{\partial x} = \cos x + \cos y + 1$

This is exact.

Scanned with OKEN Scanner

Solution,

$$\int M\,dx + \int (\text{Terms of } N \overset{\text{not}}{\hat{}} \text{ containing } x)\,dy = C$$

$$\int (y\cos x + \sin y + y)\,dx + \int 0\,dy = C$$

$$\Rightarrow \boxed{y\sin x + x\sin y + xy = C}$$

iv) $\left(y^2 e^{xy^2} + 4x^3\right)dx + \left(2xy\,e^{xy^2} - 3y^2\right)dy = 0$

→ Given: $M\,dx + N\,dy$

$M = y^2 e^{xy^2} + 4x^3$; $N = 2xy\,e^{xy^2} - 3y^2$

$\frac{\partial M}{\partial y} = e^{xy^2}2y + y^2 e^{xy^2}\cdot 2y$ $\quad \frac{\partial N}{\partial x} = 2y\left(xye^{xy^2} + e^{xy^2}\right)$

$$\boxed{\frac{\partial M}{\partial y} = 2xy^3 e^{xy^2} + 2ye^{xy^2}}$$

$\Rightarrow 2xy^3 e^{xy^2} + 2ye^{xy^2}$

$\frac{\partial M}{\partial y} = \frac{\partial N}{\partial x} \Rightarrow$ This is exact.

Solution is,

$$\int M\,dx + \int (\text{Term of } N \text{ not containing } x)\,dy = C$$

$$\int (y^2 e^{xy^2} + 4x^2)\,dx + \int -3y^2\,dy = C$$

$$y^2\frac{e^{xy^2}}{y^2} + \frac{4x^4}{4} - y^3 = C$$

$$\Rightarrow e^{xy^2} + x^4 - y^3 = C$$

Scanned with OKEN Scanner

v) $ye^{xy}\,dx + (xe^{xy} + 2y)\,dy = 0$

→ This is of the form $M\,dx + N\,dy = 0$

$M = ye^{xy}$ $\qquad N = xe^{xy} + 2y$

$\frac{\partial M}{\partial y} = yxe^{xy} + e^{xy}$ $\qquad \frac{\partial N}{\partial x} = xye^{xy} + e^{xy}$

This function is exact.

Solution is $\int M\,dx + \int$ (Terms of N not containing x) $dy = c$

$\int ye^{xy}\,dx + \int 2y\,dy = c$

$e^{xy} + y^2 = c$

vi) $(3x^2 + 6xy^2)\,dx + (6x^2y + 4y^2)\,dy = 0$

→ This is of the form $M\,dx + N\,dy = 0$

$M = 3x^2 + 6xy^2$ $\qquad N = 6x^2y + 4y^2$

$\frac{\partial M}{\partial y} = 12xy$ $\qquad = \qquad \frac{\partial N}{\partial x} = 12xy$

This function is exact.

Solution is $\int M\,dx + \int$ Terms of N not containing x dy

$\int (3x^2 + 6xy^2)\,dx + \int 4y^2\,dy = c$

$x^3 + 3x^2y^2 + \frac{4y^3}{3} = c$

Scanned with OKEN Scanner

~~$(3x^2+6xy^2)\,dx + (6x^2y+4y^2)\,dy = 0$~~

viii) $y\sin(2x)\,dx - (1+y^2+\cos^2 x)\,dy = 0$

→ $M\,dx + N\,dy = 0$

$M = y\sin(2x)$ $\quad N = -(1+y^2+\cos^2 x)$

$\frac{\partial M}{\partial y} = \sin 2x$ $\quad \frac{\partial N}{\partial x} = -(2\cos x(-\sin x))$

$= 2\cos x \sin x$

$= \sin 2x$

Sol:

$\int M\,dx + \int (N \text{ terms not containing } x)\,dy = C$

$-y\frac{\cos 2x}{2} + y + \frac{y^3}{3} = C$

LINEAR DIFFERENTIAL EQUATION:

- A D.E. is said to be linear if the dependent variable and its derivatives occur only in the first degree and not multiplied together.

Eg)

$x^2y' + 2e^x y = \cos x$ → linear

$y'y = e^{2x}$ → Non-linear

- The standard form of a linear equation of first order is

$$y' + Py = Q$$

Integrating factor: $e^{\int P\,dx}$

Sol: $y(I.F) = \int Q(I.F)\,dx + c$

P, Q → functions of x

Note: Linear D.E. in x:

$$\frac{dx}{dy} + Px = Q$$

P & Q → functions of y.

$$I.F = e^{\int P\,dy}$$

$$\text{Sol}^n : x(I.F) = \int Q\,(I.F)\,dy + C$$

Q) Solve following D.E.

i) $(x+1)\frac{dy}{dx} - y = e^{3x}(x+1)^2$

$$\rightarrow \frac{dy}{dx} - \frac{y}{(x+1)} = e^{3x}(x+1)$$

This is of the form $y' + Py = Q$

$$P = \frac{-1}{x+1} \qquad Q = e^{3x}(x+1)$$

let $x+1 = t$

$$1 = \frac{dt}{dx}$$

$$dx = dt$$

$$e^{-\int \frac{1}{t}dt} = e^{-\log t} = \frac{1}{t}$$

$$\frac{1}{x+1}\,y = \int \frac{e^{3x}}{(x+1)}(x+1)$$

$$\frac{1}{x+1} \cdot y = \frac{e^{3x}}{3} + C$$

$$y = \frac{e^{3x}(x+1) + 3C(x+1)}{3}$$

Scanned with OKEN Scanner

ii) $\cos^2 x \frac{dy}{dx} + y = \tan x$

$\rightarrow \frac{dy}{dx} + \sec^2 y = \frac{\tan x}{\cos^2 x}$

$\frac{dy}{dx} + \sec^2 y = \frac{\sin x}{\cos^3 x}$

$I.F = e^{\int P dx} = e^{\int \sec^2 x\, dx} = e^{\tan x}$

$e^{\tan x} \cdot y = \frac{\tan x}{\cos^2 x} \cdot \frac{\sin x}{\cos^3 x} \cdot e^{\tan x} + C$

$e^{\tan x}\, y = \int \tan x \sec^2 x\, e^{\tan x} dx + C$

$e^{\tan x}\, y = \int t e^t dt + C$

$e^{\tan x}\, y = te^t - \int e^t + C$

$e^{\tan x}\, y = \tan x\, e^{\tan x} - e^{\tan x} + C$

$\boxed{y = \tan x - 1 + \frac{C}{e^{\tan x}}}$ Gl. Sol^n

$\tan x = t$

$\sec^2 x\, dx = dt$

ILATE

iii) $2y' \cos x + 4y \sin x = \sin 2x$ [Given : $y = 0$ when $x = \pi/3$]

$\Rightarrow$

$y' + 2\tan x\, y = \sin x$

$P = 2\tan x \qquad Q = 2\sin x$

$I.F = e^{\int 2\tan x} = e^{2(\log(\sec x))} = \sec^2 x$

$y\sec^2 x = \int \sin x \sec^2 x + C$

$y\sec^2 x = \int \tan x \sec x + C$

$y\sec^2 x = \sec x + C$ → General solution

Sub $y=0$ & $x=\pi/3$

$0 = 2 + C \Rightarrow \boxed{C = -2}$

$\cos\frac{\pi}{3} = \frac{1}{2}$

$\sec\pi/3 = 2$

$\boxed{y\sec^2 x = \sec x - 2}$

$\boxed{y = \cos x - 2\cos^2 x}$ → Particular Solution

$(x + 2y^3)\frac{dy}{dx} = y$

This is linear in x — This is not linear in y.

$\frac{dy}{dx} = \frac{y}{x+2y^3}$

$\frac{x}{y} + 2y^2 = \frac{dx}{dy}$

$\frac{dx}{dy} - \frac{x}{y} = 2y^2$

$P = \frac{-1}{y}$ $\quad Q = 2y^2$

I.F: $e^{\int \frac{-1}{y} dy} = e^{-\log y} = \frac{1}{y}$

$\frac{x}{y} = \int \frac{2y^2}{y}$

$\frac{x}{y} = \int 2y\, dy$

$\boxed{\frac{x}{y} = y^2 + C}$

$\boxed{x = y^3 + Cy}$

Scanned with OKEN Scanner

BERNOULLI'S EQUATIONS :

- Equation of the form $\boxed{\frac{dy}{dx} + Py = y^n Q}$ where P, Q are functions of x is called a Bernoulli's Equation.
- This can be reduced to a linear form and solved accordingly.

WORKING RULE :

* Consider D.E. $\boxed{dy/dx + Py = y^n Q}$

i) Divide by $y^n \Rightarrow$ we get $\boxed{y^{-n}\frac{dy}{dx} + Py^{1-n} = Q}$ ... (1)

ii) Put $\boxed{y^{1-n} = z}$

iii) Diff. w.r.t $x \Rightarrow (1-n)\, y^{-n}\frac{dy}{dx} = \frac{dz}{dx}$

$$\Rightarrow \boxed{y^{-n}\frac{dy}{dx} = \frac{1}{(1-n)}\frac{dz}{dx}}$$

iv) Sub this in eq (1). $\Rightarrow$

$$\frac{1}{(1-n)}\frac{dz}{dx} + Py^{1-n} = Q \quad \frac{1}{(1-n)}\frac{dz}{dx} + Pz = Q$$

$$\Rightarrow \boxed{\frac{dz}{dx} + (1-n)Pz = Q(1-n)}$$

This is the linear D.E. in 'z' & can be solved as discussed previously.

Scanned with OKEN Scanner

Q) Solve the following D.Eq:

i) $2\frac{dy}{dx} + y = x^3 y^6$

$\Rightarrow \frac{dy}{dx} + \frac{y}{x} = x^2 y^6$

This is of the form $\frac{dy}{dx} + Py = Q y^n$ $\Rightarrow$ Bernoulli's eqn.

$\div y^6 \Rightarrow y^{-6}\frac{dy}{dx} + \frac{y^{-5}}{x} = x^2 \rightarrow (1)$

Put $y^{-5} = z$

$-5y^{-6}\frac{dy}{dx} = \frac{dz}{dx}$

$y^{-6}\frac{dy}{dx} = \frac{-1}{5}\frac{dz}{dx}$

Sub in (1)

$\Rightarrow \frac{-1}{5}\frac{dz}{dx} + \frac{y^{-5}}{x} = x^2$

$\Rightarrow \boxed{\frac{dz}{dx} - \frac{5}{x}z = -5x^2}$

This is linear in z. $[P = -5/x, \ Q = -5x^2]$

I.F $= e^{\int P dx} = e^{\int -5/x\, dx} \Rightarrow e^{-5\log x\, dx}$

$\Rightarrow \frac{1}{x^5}$.

Solution $\Rightarrow \frac{z}{x^5} = \int \frac{-5x^2}{x^5} + C$

x^{-3}

$\frac{x^{-2}}{-2}$

$\frac{z}{x^5} = \int \frac{-5}{x^3} + C \Rightarrow \frac{z}{x^5} = \frac{5}{2}x^{-2} + C$

$\Rightarrow \boxed{z = \frac{5}{2}x^3 + Cx^5}$

$y^{-5} = \frac{5}{2}x^3 + Cx^5$

ii) $\frac{dy}{dx} = y\tan x - y^2 \sec x$

$\Rightarrow$ Given:

$$\frac{dy}{dx} - y\tan x = -y^2 \sec x$$

This is a Bernoulli's Eqn.

$\div y^2$

$$\Rightarrow y^{-2}\frac{dy}{dx} - \frac{1}{y}\tan x = -\sec x$$

$$\Rightarrow \boxed{y^{-2}\frac{dy}{dx} - y^{-1}\tan x = -\sec x}$$

let $y^{-1} = z$

$$\frac{-1}{y^2}\frac{dy}{dx} = \frac{dz}{dx}$$

$$\Rightarrow \boxed{y^{-2}\frac{dy}{dx} = -\frac{dz}{dx}}$$

$$\Rightarrow \frac{-dz}{dx} - z\tan x = -\sec x$$

$$\Rightarrow \frac{dz}{dx} + \tan x\,(z) = \sec x$$

This is a linear D.E in z

$P = \tan x$, $Q = \sec x$

$$I.F = e^{\int P\,dx} = e^{\int \tan x} = e^{\log(\sec x)} = \sec x$$

$$z(\sec x) = \int \sec^2 x + C$$

$$\sec x = \tan x + C$$

$$\Rightarrow y^{-1} = \sin x + C[\cos x]$$

Scanned with OKEN Scanner

iii) $x\frac{dy}{dx} + y = -2x^6y^4$

$\Rightarrow \frac{dy}{dx} + \frac{y}{x} = -2x^5y^4$.

$\div y^4 \Rightarrow y^{-4}\frac{dy}{dx} + \frac{y^{-3}}{x} = -2x^5$

let $z = y^{-3}$

$\frac{dz}{dx} = -3y^{-4}\frac{dy}{dx}$

$\Rightarrow y^{-4}\frac{dy}{dx} = \frac{-1}{3}\frac{dz}{dx}$

$\frac{-1}{3}\frac{dz}{dx} + \frac{z}{x} = -2x^5$

$\frac{dz}{dx} - \frac{3z}{x} = 6x^5$

$e^{\int -\frac{3}{x}dx} = e^{-3\log x} = \frac{1}{x^3}$

$\frac{z}{x^3} = \int 6x^5 e^{-3x^2/2} + c$

$z e^{-3x^2/2} = 6\int x^5 e^{-3x^2/2} + c$

$\frac{z}{x^3} = 6\int \frac{x^5}{x^3} + c$

$\frac{z}{x^3} = 6\frac{x^3}{3} + c$

$\frac{z}{x^3} = 2x^3 + c$

$\boxed{z = 2x^6 + cx^3}$

$\therefore y^{-3} = 2x^6 + cx^3$ //

Scanned with OKEN Scanner

iv) $(x^3y^2 + xy)\,dx = dy$

$\rightarrow x^3y^2 + xy = \frac{dy}{dx}$

$\Rightarrow \frac{dy}{dx} - xy = x^3y^2$

$\div y^2 \Rightarrow y^{-2}\frac{dy}{dx} - xy^{-1} = x^3$

let $y^{-1} = z$

$-y^{-2} = \frac{dz}{dx}$

$$\boxed{y^{-2}\frac{dy}{dx} = -\frac{dz}{dx}}$$

$\frac{-dz}{dx} - xz = x^3$

$\frac{dz}{dx} + xz = -x^3$

$e^{\int x\,dx} = e^{x}$

$ze^x = -\int x^3e^x + c$

$x^3e^x - \int 3x^2e^x$

$x^3e^x - 3[x^2e^x - \dots]$

x^3e^x

v) $2xy' = 10x^3y^5 + xy$

vi) $x^3\frac{dy}{dx} - x^2y = -y^4\cos x$

v) $2xy' = 10x^3y^5 + xy$

$y' = 5x^2y^5 + \frac{1}{2}y$

$\frac{dy}{dx} - \frac{y}{2} = 5x^2y^5$

$y^{-5}\frac{dy}{dx} - \frac{y^{-4}}{2} = 5x^2$

let $y^{-4} = z$

$-4y^{-5}\frac{dy}{dx} = \frac{dz}{dx}$

$$y^{-5}\frac{dy}{dx} = \frac{-1}{4}\frac{dz}{dx}$$

$\frac{-1}{4}\frac{dz}{dx} - \frac{z}{2} = 5x^2$

$\frac{dz}{dx} + 2z = -20x^2$

$e^{\int 2} = e^{2x}$

$ze^{2x} = \int -20x^2e^{2x}$

$ze^{2x} = -20\left[\frac{x^2e^{2x}}{2} - \int xe^{2x}\right]$

$= -20\left[\frac{x^2e^{2x}}{2} - \left[\frac{xe^{2x}}{2} - \frac{e^{2x}}{2}\right]\right]$

$= -10[x^2e^{2x} - xe^{2x} - e^{2x}] \Rightarrow -10e^{2x}[x^2 - x - 1]$

Scanned with OKEN Scanner

vi) $x^3\frac{dy}{dx} - x^2y = -y^4\cos x$

$\rightarrow \frac{dy}{dx} - \frac{y}{x} = -y^4 x^{-3}\cos x$

$\div y^4 \Rightarrow y^{-4}\frac{dy}{dx} - \frac{y^{-3}}{x} = -x^{-3}\cos x$

let $y^{-3} = z$

$-3y^{-4}\frac{dy}{dx} = \frac{dz}{dx}$

$y^{-4}\frac{dy}{dx} = \frac{-1}{3}\frac{dz}{dx}$

$\frac{-1}{3}\frac{dz}{dx} - \frac{z}{x} = -x^{-3}\cos x$

$\frac{dz}{dx} + \frac{3z}{x} = 3x^{-3}\cos x$

I.F $= e^{3\int\frac{1}{x}} = e^{+3\log x} \Rightarrow x^3$

$zx^3 = \int 3x^3 . x^{-3}\cos x + c$

$zx^3 = \int 3\cos x + c$

$zx^3 = 3\sin x + c$

$y^{-3}x^3 = 3\sin x + c$

2nd Higher Order Linear D.E.

The general form of a L.D.E of n^{th} order with constant coefficient is,

$$\frac{d^n y}{dx^n} + a_1 \frac{d^{n-1} y}{dx^{n-1}} + \dots + a_{n-1} \frac{dy}{dx} + a_n y = \phi(x)$$

$a_1, a_2 \Rightarrow$ constants

$\phi(x) \Rightarrow$ function of x.

OPERATOR D:

"D" is known as differential operator of $y = f(x)$, then we write

$$\frac{dy}{dx} = Dy \quad \left(D \Rightarrow \frac{d}{dx}\right)$$

$$\frac{d^2 y}{dx^2} = D^2 y$$

$\Rightarrow$ Can be rewritten as

$$(D^n + a_1 D^{n-1} + a_2 D^{n-2} + \dots + a_{n-1} D + a_n) y = \phi(x).$$

$$\Rightarrow f(D) y = \phi(x).$$

Solution of D.E. of $f(D)y = \phi(x)$:-

Solution of L.D.E. $f(D)y = \phi(x)$ is

$$\boxed{y = y_c + y_p}$$

$y_c \Rightarrow$ complementation function & is a soln of $f(D) y = 0$.

y_p ⇒ particular integral & is a soln of $f(D)y = \phi(x)$.

Eg) $y'' + y = x^2$

$(D^2+1)y = x^2$

$y = \underbrace{c_1 \cos x + c_2 \sin x}_{y_c} + \underbrace{x^2 - 2}_{y_p}$

$y_c = c_1 \cos x + c_2 \sin x$

$y_c'' = -c_1 \cos x - c_2 \sin x$

$y_c'' + y_c = 0$

$y_p = x^2 - 2$

$y_p'' = 2$

$y_p'' + y_p = x^2$

Rules for finding Complementary Solution

Consider the homogeneous equation

$f(D)y = 0$

The equation $f(m)=0$ obtained by replacing D by m in $f(D)y$ is known as Auxiliary equation of given differential equation.

$y''' + y'' + y' = 0$

$(D^3 + 2D^2 + D)y' = 0$

$m^3 + 2m^2 + m + 1 = 0$] Auxiliary Eqn (A.E.)

Scanned with OKEN Scanner

PAGE NO.
DATE / /

Let m_1, m_2, m_3 be the roots of above equation, we obtain y_c as per given table.

Nature of roots	Complementary Function
1) Real, distinct m_1, m_2, m_3	$y_c = C_1 e^{m_1 x} + C_2 e^{m_2 x} + C_3 e^{m_3 x}$
2) Real and equal $m_1 = m_2$ (m_3 is distinct)	$y_c = (C_1 + C_2 x) e^{m_1 x} + C_3 e^{m_3 x}$
3) Real & equal $m_1 = m_2 = m_3$	$y_c = (C_1 + C_2 x + C_3 x^2) e^{m_1 x}$
4) A pair of imaginary roots $m_1, m_2 = \alpha \pm i\beta$	$y_c = e^{\alpha x}(C_1 \cos\beta x + C_2 \sin\beta x)$

Q) Solve the following D.E.

i) $y'' + y' - 2y = 0$

$\rightarrow (D^2 + D - 2) y = 0$

To find y_c :-

A.E $\Rightarrow m^2 + m - 2 = 0$

$m = 1 \; ; \; m = -2.$

$$\boxed{y_c = C_1 e^{x} + C_2 e^{-2x}}$$

$\therefore$ Soln of given D.E $\Rightarrow y = C_1 e^{x} + C_2 e^{-2x}$ //

Scanned with OKEN Scanner

Q) $y'' + 10y' + 25y = 0$

→ $(D^2 + 10D + 25)y = 0$

To find y_c

A.E: $m^2 + 10m + 25 = 0$

$m = -5, -5$

$\boxed{m_1 = m_2 = -5}$

$y_c = (C_1 + C_2 x)e^{-5x}$

Q) $(D^2 - 6D + 13)y = 0$

→ To find y_c.

A.E :- $m^2 - 6m + 13 = 0$

$m_1 = 3+2i, m_2 = 3-2i \Rightarrow m = 3 \pm 2i$

$\alpha = 3, \beta = 2$

$y_c = e^{3x}\{C_1 \cos 2x + C_2 \sin 2x\}$

$e^{\alpha x}(C_1 \cos \beta x + C_2 \sin$

Q) $y''' - 2y'' + 4y' - 8y = 0$

→ $(D^3 - 2D^2 + 4D - 8)y = 0$

To find y_c:

A.E: $m^3 - 2m^2 + 4m - 8 = 0$

$m_1 = 2; m_2 = 2i; m_3 = -2i$

Scanned with OKEN Scanner

PAGE NO.
DATE / /

$m = 2, \pm 2i \qquad (\alpha = 0, \beta = 2)$

$y_c = c_1 e^{2x} + e^{0x}\{c_2 \cos 2x + c_3 \sin 2x\}$.

General soln : $y = c_1 e^{2x} + c_2 \cos 2x + c_3 \sin 2x$.

Q) $(D^3 + 1)y = 0$

To find y_c. => A.E

→ $m^3 + 1 = 0$

$m_1 = -1, \cancel{m_2 = 0.5, \quad m_3 = 0.5}$

$m_2 = 0.5 + 0.866i$

$m_3 = 0.5 - 0.866i$

$\Rightarrow m_2 = 0.5 \pm 0.866i$

$y_c = c_1 e^{-x} + e^{0.5x}\{c_2 \cos(0.866x) + c_3 \sin(0.866x)\}$.

Q) $\cancel{D^3}(D^4 - 5D^2 + 4)y = 0$

→ To find y_c.

A.E. $= m^4 - 5m^2 + 4 = 0$ $\qquad n^2 - 5n + 4 = 0$

$m_1 = \pm 2$ $\qquad m_1^2 = 4$

$m_2 = \pm 1$ $\qquad m_2^2 = 1$

$y_c = c_1 e^x + c_2 e^{-x} + c_3 e^{2x} + c_4 e^{-2x}$.

Obtain Particular Soln of D.E.

i) $(D^3 - 2D^2)y = 0$ Given that $y(0) = -3$,

$y'(0) = 0$

$y''(0) = 12$

→

To find y_c.

A.E: $m^3 - 2m^2 = 0$

$\Rightarrow m^2(m-2) = 0$

$m = 0, 0, 2$

$\Rightarrow m_1 = 0 \; ; \; m_2 = 2$

$y_c = (C_1 + C_2 x)e^{0x} + C_3 e^{2x}$

$y_c = (C_1 + C_2 x) + C_3 e^{2x}$

General Solution $\Rightarrow$ $\boxed{y_c = C_1 + C_2 x + C_3 e^{2x}}$

To find particular solution:

Given $y(0) = -3$, $y'(0) = 0$, $y''(0) = 12$.

$y(0) = C_1 + C_3 e^0$

$y(0) = \boxed{C_1 + C_3 = -3}$

$y'(x) = C_2 + 2 \cdot C_3 e^{2x}$

$0 = C_2 + 2C_3$

$\boxed{C_2 = -2C_3}$

$y'' = 4C_3 e^{2x}$

$y''(0) = \boxed{4C_3 = 12}$

$\boxed{C_3 = 3}$

$\boxed{C_2 = -6}$

$\boxed{C_1 = -6}$

$y = -6 - 6x + 3e^{2x}$ [Particular Solution]

Scanned with OKEN Scanner

PAGE NO.
DATE / /

ii) $\frac{d^2x}{dt^2} - 4\frac{dx}{dt} + 13x = 0$ & $x(0) = \frac{dx(0)}{dt} = 2$

$x(0) = x'(0) = 2$

→ $D = \frac{d}{dt}$

$(D^2 - 4D)(D^2 - 4D + 13)x = 0$

To find y_c:

A.E :- $m^2 - 4m + 13 = 0$

$m = 2 + 3i, 2 - 3i$

$m = 2 \pm 3i$ $(\alpha = 2 ; \beta = 3)$

$x_c = e^{2t}(C_1 \cos 3t + C_2 \sin 3t)$

To find Particular Solution.

$x = e^{2t}(C_1 \cos 3t + C_2 \sin 3t)$

$x(0) = e^0(C_1 \cos 0 + C_2 \sin 0)$

$x(0) = C_1 = 2$

$x' = 2e^{2t}[C_1 \cos 3t + C_2 \sin 3t] + e^{2t}[-3C_1 \sin 3t + 3C_2 \cos 3t]$

$x'(0) = 2e^0[2\cos 0] + e^0[3C_2 \cos 0]$

$x'(0) = 4 + 3C_2 = 2$

$3C_2 = -2$

$C_2 = -2/3$

∴ Particular Solution ⇒

$x = e^{2t}\left(2\cos 3t - \frac{2}{3}\sin 3t\right)$

Scanned with OKEN Scanner

iii) $y'' + 2y' + 10y = 0$

Given $y(0) = 4$; $y'(0) = 1$

$\rightarrow (D^2 + 2D + 10)y = 0$

To find y_c:

$m^2 + 2m + 10 = 0$

$m = -1 \pm 3i$ $\quad \alpha = -1, \beta = 3$.

$y_c = e^{-x}(C_1 \cos 3x + C_2 \sin 3x) \Rightarrow y = e^{-x}(C_1 \cos 3x + C_2 \sin 3x$

For particular Solution.

$y = e^{-0}(C_1 \cos 0) = 4 = C_1$.

$y' = -e^{-x}(C_1 \cos 3x + C_2 \sin 3x) + 3e^{-x}(-C_1 \sin 3x + C_2 \cos 3x$

$1 = -(C_1) + 3(C_2)$

$1 = -4 + 3C_2$

$\frac{5}{3} = C_2$.

$y = e^{-x}\left(4 \cos 3x + \frac{5}{3} \sin 3x\right)$.

Scanned with OKEN Scanner

INVERSE DIFFERENTIAL OPERATOR

- The Inverse Differential Operator is denoted by $\frac{1}{D}$.
- If $\phi(x)$ is any function of x, $\frac{1}{D}\phi(x) = \int \phi(x)\,dx$

Particular Integral :

- Consider a non-homogeneous L.D.E of $f(D)y = \phi(x)$
The General Sol^n of this is given by $\boxed{y = y_c + y_p}$

$$\boxed{y_p = \frac{1}{f(D)}\phi(x)}$$

$y = y_c + y_p$

$f(D)y = f(D)[y_c + y_p] = \phi(x)$

$= f(D)y_c + f(D)y_p = \phi(x)$

$= f(D)y_p = \phi(x)$

$\Rightarrow \boxed{y_p = \frac{\phi(x)}{f(D)}}$

Rules of finding y_p :

Case 1: When $\phi(x) = e^{ax}$

i) $y_p = \frac{1}{f(D)}e^{ax} = \frac{1}{f(a)}e^{ax}$; when $f(a) \neq 0$ replace D by a

ii) if $f(a) = 0$, $y_p = x \cdot \frac{1}{f'(a)}e^{ax}$; when $f'(a) \neq 0$

iii) If $f'(a) = f''(a) = 0$, $y_p = x^2 \cdot \frac{1}{f''(a)}e^{ax}$; when $f''(a) \neq 0$ & so on.

Q) Find General Sol^n of following D.E.

i) $y'' - 3y' + 2y = e^{3x}$

$\rightarrow (D^2 - 3D + 2)y = e^{3x}$

* To find y_c :

A.E : $m^2 - 3m + 2 = 0$

$m = 1, 2$

$y_c = C_1 e^x + C_2 e^{2x}$

* To find y_p :

$\phi(x) = e^{3x}$

$y_p = \frac{1}{f(D)} \phi(x) = \frac{1}{f(D)} e^{3x}$ (Replace D by a (a = 3))

$f(D) = D^2 - 3D + 2$

$f(3) = 9 - 9 + 2 = 2$

$y_p = \frac{1}{f(3)} e^{3x} = \frac{1}{2} e^{3x} = \frac{e^{3x}}{2}$

∴ General Solution is,

$y = y_c + y_p$

$y = C_1 e^x + C_2 e^{2x} + \frac{e^{3x}}{2}$

ii] $(D^2 - 3D + 4) y = 12 e^{3x}$

→ To find y_c : 1 x 4

Auxiliary Equation:

$m^2 - 3m + 4$

$m = 1.5 \pm 1.32i = \frac{3}{2} \pm \frac{\sqrt{7}}{2} i$

$\alpha = \frac{3}{2}$; $\beta = \frac{\sqrt{7}}{2}$

Scanned with OKEN Scanner

$y_c = e^{3/2x}\{C_1 \cos\frac{\sqrt{7}}{2}x + C_2 \sin\frac{\sqrt{7}}{2}x\}$

To find y_p:

$y_p = \frac{1}{f(D)}\phi(x)$

$\frac{1}{f(0)}\, 12e^{2x}$ {Replace $D = a = 2$}

$f(D) = f(a) = f(2) = D^2 - 3D + 4$

$= 4 - 6 + 4$

$= 2$.

$y_p = \frac{12e^{2x}}{2} = 6e^{2x}$

General Soln:

$y = y_c + y_p$

$= e^{3/2x}\{C_1 \cos\frac{\sqrt{7}}{2}x + C_2 \sin\frac{\sqrt{7}}{2}x\} + 6e^{2x}$

iii) $y'' + 4y' + 4y = e^{-2x}$

$\rightarrow (D^2 + 4D + 4)y = e^{-2x}$

To find y_c:

$m^2 + 4m + 4 = 0$

$m = -2, -2$

-2×-2

$y_c = (C_1 + C_2 x)e^{-2x}$

To find y_p:

$y_p = \frac{1}{f(D)}\phi(x)$

$= \frac{1}{f(D)} e^{-2x}$

Replace D by $a = -2$

$f(D) = D^2 + 4D + 4$

$f(a) = f(-2) = 4 - 8 + 4 = 0$.

Scanned with OKEN Scanner

Since $f(a) = 0$

$f'(D) = 2D + 4$

$f'(a) = 2(-2) + 4 = 0$.

$f''(D) = 2$ $\quad$ $f''(-2) = 2$

$y_p = x^2 \cdot \frac{1}{f''(a)} \phi(x)$

$= x^2 \cdot \frac{1}{2} e^{-2x} = \frac{e^{-2x} x^2}{2}$.

~~General~~

$y = y_c + y_p$

$= (C_1 + C_2 x) e^{-2x} + \frac{x^2 e^{-2x}}{2}$.

[iv] $(D-3)^2 y = e^x + 2e^{3x}$

→ To find y_c:

A.E :- $(m-3)^2 = 0$

$\Rightarrow \boxed{m = 3, 3}$

$y_c = (C_1 + C_2 x) e^{3x}$

To find y_p:

$y_p = \frac{1}{f(D)} \phi(x)$

$y_p = \frac{1}{f(D)} (e^x + 2e^{3x})$

$y_p = \frac{1}{f(D)} (e^x) + \frac{1}{f(D)} (2e^{3x})$

I) Replace D by 1

$f(D) = (D-3)^2$

$f(1) = (1-3)^2 = 4$.

II) Replace D by 3.

$f(D) = f(3) = 0$.

$f'(D) = 2(D-3) = /$ $f''(D) = 2$.

$f'(3) =$

$$y_p = \frac{1}{4}(e^x) + \frac{1}{2}(2e^{3x})$$

$$y_p = \frac{1}{4}e^x + e^{3x}$$

$$y = y_c + y_p$$

$$= (c_1 + c_2 x)e^{3x} + \frac{1}{4}e^x + e^{3x}$$

V $(4D^2 - 1)y = e^{x/2} + 12e^x + 4$

→ To find y_c:

A.E $= 4m^2 - 1 = 0$

$$m = \frac{1}{2}, \frac{-1}{2}$$

$$y_c = c_1 e^{1/2 x} + c_2 e^{-1/2 x}$$

To find y_p:

$$y_p = \frac{1}{f(D)}\phi(x)$$

$$= \frac{1}{f(D)}(e^{x/2} + 12e^x + 4e^{0x})$$

$$= \frac{1}{f(D)}e^{x/2} + \frac{1}{f(D)}12e^x + \frac{1}{f(D)}4e^{0x}$$

Replace D by $\frac{1}{2}$

$f(D) = 4D^2 - 1$

$f(\frac{1}{2}) = 0$.

$f'(D) = 8D$

$f'(\frac{1}{2}) = 4$.

Replace D by 1

$f(D) = 4D^2 - 1$

$f(1) = 3$.

Replace D by 0

$f(D) = 4D^2 - 1$

$f(0) = -1$.

$$y_p = x \cdot \frac{1}{4}e^{x/2} + \frac{1}{3}12e^x + 4e^{2x}$$

$$y_p = \frac{x}{4}e^{x/2} + 4e^x - 4e^{2x}$$

$$y = c_1 e^{1/2 x} + c_2 e^{-1/2 x} + \frac{x}{4}e^{x/2} + 4e^x - 4e^{2x}$$

Scanned with OKEN Scanner

vi) $y''' - 6y'' + 11y' - 6y = e^{-2x} + e^{+3x}$

$\Rightarrow$ ~~To find~~ $(D^3 - 6D^2 + 11D - 6)\, y = e^{+2x} + e^{+3x}$

To find y_c:

AE: $m^3 - 6m^2 + 11m - 6$

$\Rightarrow m = 1, 3, 2$

$y = C_1 e^{x} + C_2 e^{2x} + C_3 e^{3x}$

To find y_p:

$y_p = \frac{1}{f(D)} \phi(x) \Rightarrow y_p = \frac{1}{f(D)} (e^{2x} + e^{3x})$

$\Rightarrow y_p = \frac{1}{f(D)} e^{2x} + \frac{1}{f(D)} e^{3x}$

I: $a = 2 \quad D^3 - 6D^2 + 11D - 6$

$f(D)\ 8 - 24 + 22 - 6 = 0$

$f'(D) = 3D^2 - 12D + 11$

$f'(2) = 12 - 24 + 11 = -1$

$y_p = -x e^{2x}$

II: $a = 3, \quad D^3 - 6D^2 + 11D - 6$

$f(D) = 27 - 54 + 33 - 6$

$= 0$

$f'(D) = 3D^2 + 12D + 11$

$= 27 - 36 + 11$

$= 2$

$y_p = 2\, x/2\, e^{3x}$

$y_p = -x e^{2x} + \frac{x}{2} e^{3x}$

$y = y_c + y_p$

$= C_1 e^{x} + C_2 e^{2x} + C_3 e^{3x} - x e^{2x} + \frac{x}{2} e^{3x}$

Scanned with OKEN Scanner

PAGE NO.
DATE / /

Case II) When $\phi(x) = \sin(ax)$ (or $\cos(ax)$)

$$y_p = \frac{1}{f(D^2)} \sin(ax)$$

i.e.,

i) $y_p = \frac{1}{f(-a^2)} \sin(ax)$; $f(-a^2) \neq 0$ {Replace D^2 by $-a^2$}

ii) if $f(-a^2) = 0$, then $y_p = x \cdot \frac{1}{f'(-a^2)} \sin(ax)$; $f'(-a^2) \neq 0$

Q) Solve the following D.E :

i) $y'' + 16y = \sin t$ ~~$\sin t$~~ $14\cos(3x)$

To find y_c

→ ~~the~~ $= (D^2 + 16)\, y = 0$

A.E : $m^2 + 16 = 0$

$m = \pm 4i$

$\alpha = 0, \beta = 4$

$y_c = e^{0x}(C_1 \cos 4x + C_2 \sin 4x) = C_1 \cos 4x + C_2 \sin 4x$

$y_p = \frac{1}{f(D^2)} \phi(x) = \frac{1}{f(D^2)} 14\cos 3x$

$f(D^2) = D^2 + 16$

D^2 by $-a^2 = -9$

$f(-9) = -9 + 16 = 7$.

$y_p = \frac{1}{7} 14\cos 3x$

$y_p = 2\cos 3x$

$y = C_1 \cos 4x + C_2 \sin 4x + 2\cos 3x$ //

Scanned with OKEN Scanner

ii) $(D^2+36)y = 4\cos(6x)$

→ To find y_c:

A.E: $m^2+36=0$

$m = \pm 6i$

$\alpha = 0 ; \beta = 6$

$y_c = e^{0}(c_1\cos 6x + c_2 \sin 6x)$

To find y_p:

$y_p = \frac{1}{f(D^2)}\phi(x)$ Replace D^2 by $-a^2 = -36$

$y_p = \frac{1}{f(D^2)} 4\cos(6x)$

$f(D^2) = D^2+36$

$= -a^2+36$

$= -36+36 = 0.$

$f'(D) = 2D$

~~$f''(D^2) = 2\times 36 = 72$~~.

~~$y_p = x\cdot\frac{1}{72}\cdot 4\cos(6x) = \frac{x\cos(6x)}{18}$~~.

$\therefore y_p = \frac{x}{2D}\cdot 4\cos 6x = \frac{2x\cos(6x)}{D} = 2x\int\cos 6x$

$y_p = 2x\frac{\sin 6x}{6} = \frac{x\sin 6x}{3}$.

∴ The General Solution is $y = y_c + y_p$

$y = c_1\cos 6x + c_2\sin 6x + \frac{x\sin 6x}{3}$.

Scanned with OKEN Scanner

PAGE NO.
DATE / /

or $y_p = x \cdot \frac{1}{20} \cos(6x)$

multiply & Divide by D.

$$y_p = 2x \cdot \frac{D}{D^2} (\cos 6x)$$

$$\Rightarrow y_p = \frac{2x}{-36} \frac{d}{dx} (\cos 6x)$$

$$y_p = \frac{x \sin x}{3}$$

iii) $(D^3 + D^2 + D + 1)\, y = \cos 2x$

→ To find y_c:

A.E: $m^3 + m^2 + m + 1 = 0$

$m = -1, \pm i$, $\quad \alpha = 0, \beta = 1$

$$y_c = C_1 e^{-x} + C_1 \cos x + C_2 \sin x$$

To find y_p:

$$y_p = \frac{1}{f(D)} \phi(x)$$

$$= \frac{1}{f(D)} \cos(2x) \qquad \text{replace } D \text{ by } -a^2 = -4$$

$$f(D) = D^3 - 4 + D + 1$$

$$= D \cdot D^2 - 4 + D + 1 = -4D - 4 + D + 1$$

$$= -3D - 3 \Rightarrow -3(D+1)$$

$$y_p = \frac{1}{-3(D+1)} \cos(2x)$$

$$\frac{-1}{3}\frac{(D-1)}{(D+1)(D+1)}\cos(2x)$$

$$= \frac{-1}{3}\frac{D(\cos 2x) - \cos 2x}{(D^2-1)}$$

$$= \frac{-1}{3}\frac{-2\sin(2x) - \cos 2x}{-3-1} = \frac{-1}{15}(2\sin 2x + \cos 2x)$$

iii) $y = C_1\cos x + C_2\sin x - \frac{1}{15}(2\sin 2x + \cos 2x)$

iv) $y'' + 2y' + 5y = 6\sin(2x) + 7\cos(2x)$

$\rightarrow (D^2 + 2D + 5)y$

⇒ To find y_c:

A.E: $m^2 + 2m + 5 = 0$

$$m = -1 \pm 2i$$

$$\alpha = -1, \quad \beta = 2$$

$$y_c = e^{-x}(C_1\cos 2x + C_2\sin 2x)$$

To find y_p:

$$y_p = \frac{1}{f(D)}(6\sin 2x + 7\cos 2x)$$

$$= \underbrace{\frac{1}{f(D)}6\sin(2x)}_{I} + \underbrace{\frac{1}{f(D)}7\cos(2x)}_{II}$$

Replace D^2 by -4.

Scanned with OKEN Scanner

D^2+2D+5

$= -4+2D+5 = 2D+1$

I) $\frac{1}{2D+1}\ 6\sin(2x)$

$\frac{1}{(2D+1)}\frac{(2D-1)}{(2D-1)} = \frac{2D-1}{4D^2-1} = \frac{2D-1}{-17}(6\sin 2x)$

$\frac{12D(\sin 2x) - 6\sin(2x)}{-17} \Rightarrow \frac{24\cos 2x - 6\sin 2x}{-17}$

$\Rightarrow \frac{-6}{17}(\cos 2x - \sin 2x)$

II) by

$\frac{1}{2D+1}\cos 2x = \frac{1}{2D+1}\times\frac{2D-1}{2D-1}\times\cos 2x$

$= \frac{-28\sin 2x - 7\cos 2x}{-17} = \frac{7}{17}(\sin 2x - \cos 2x)$

$y = e^{-x}(C_1\cos 2x + C_2\sin 2x) - \frac{6}{17}(\cos 2x - \sin 2x) + \frac{7}{17}(\sin 2x - \cos 2x)$

$y'' + 9y = \cos 2x \cos x$

$(D^2+9)y = \cos 2x \cos x$

To find y_c:

A.E: $m^2+9=0$

$m^2 = -9$; $m = \pm 3i$. $\alpha = 0$; $\beta = 3$.

$y_c = (C_1\cos 3x + C_2\sin 3x)$

Scanned with OKEN Scanner

To find y_p:

$y_p = \frac{1}{f(D)} \phi(x)$ Replace

$y_p = \frac{1}{f(D)} (\cos 2x \cos x)$

$= \frac{1}{f(D)} \left[\frac{1}{2}(\cos 3x + \cos x)\right]$

$= \frac{1}{2}\left[\left[\frac{1}{f(D)} \cos(3x)\right] + \left[\frac{1}{f(D)} \cos(x)\right]\right]$

I II

$= \frac{1}{2}$

$\therefore f(D) = D^2 + 9$

$-9 + 9 = 0.$

$f'(D) = 2D.$

$y = \frac{x}{f'(0)} \phi(x)$ $y_p = \frac{x}{2D^2} D(\cos 3x)$

$= \frac{x}{18} 3 \sin(3x) = \frac{x \sin 3x}{6}.$

II) $\frac{1}{f(D)} \cos x$

$f(D) = -1 + 9 = 8.$

II: $\frac{1}{8} \cos x$

$y_p = \frac{1}{2}\left[\frac{x \sin 3x}{6} + \frac{\cos x}{8}\right]$

Scanned with OKEN Scanner

PAGE NO.
DATE 13/10/25

$y = C_1 \cos 3x + C_2 \sin 3x + \frac{2\sin 3x}{12} + \frac{\cos x}{16}$.

Case III) When $\phi(x) = x^m$, where m is a positive integer.

$y_p = \frac{1}{f(D)} x^m = [f(D)]^{-1} x^m$

a) Expand $[f(D)]^{-1}$ in ascending powers of D as far as the term D^m and operate on x^m term by term.

Note:

- $\frac{1}{1+t} = 1 - t + t^2 - t^3 + \dots$
- $\frac{1}{1-t} = 1 + t + t^2 + t^3 + \dots$

b) $y_p = \frac{1}{f(D)} x^m$

Use division method to obtain y_p.

Q) Solve the following D.E.

i) $y'' + 3y' + 2y = 12x^2$

→ $(D^2 + 3D + 2)y = 12x^2$

To find y_c:

AE ⇒ $m^2 + 3m + 2$

$m = -1, -2$

$y_c = C_1 e^{-x} + C_2 e^{-2x}$.

To find y_p:

$y_p = \frac{1}{f(D)} \phi(x)$

$y_p = \frac{1}{f(D)} 12x^2$

$y_p = \frac{1}{(D^2 + 3D + 2)} 12x^2$

$y_p = \frac{1}{2\left(1 + \frac{3}{2}D + \frac{1}{2}D^2\right)} \cdot 12x^2$

Scanned with OKEN Scanner

$$y_p = 6\left\{1 - \left(\frac{3}{2}D + \frac{1}{2}D^2\right) + \left(\frac{3}{2}D + \frac{1}{2}D^2\right)^2\right\}x^2$$

$$= 6\left\{1 - \frac{3}{2}D - \frac{1}{2}D^2 + \frac{9}{4}D^2\right\}x^2$$

$$= 6\left\{x^2 - \frac{3}{2}\cdot 2x - \frac{1}{2}\cdot 2 + \frac{9}{4}\cdot 2\right\}$$

$$= 6\left\{x^2 - 3x - 1 + \frac{9}{2}\right\}$$

$$y_p = 6x^2 - 18x + 21$$

$$y = C_1e^{-x} + C_2e^{-2x} + 6x^2 - 18x + 21$$

Second method:

$$2 + 3D + D^2 \;\big|\; 10x^2 \;\big|\; 6x^2 - 18x + 21$$

$$12x^2 + 36x + 12$$

$$-36x - 12$$

$$-36x - 54$$

$$42$$

$$42$$

$$0$$

$$y = C_1e^{-x} + C_2e^{-2x} + 6x^2 - 18x + 21$$

ii) $y'' + 2y' + 2y = 1 + 3x + x^2$

→

$$(D^2 + 2D + 2)y = 1 + 3x + x^2$$

To find y_c:

A.E. : $m^2 + 2m + 2 = 0$

$m = -1 \pm i$ $\quad \alpha = -1, \beta = 1$

$$y_c = e^{-x}(C_1\cos x + C_2\sin x)$$

Scanned with OKEN Scanner

~~$1+3x+x^2$~~

$$y_p = \frac{1}{(D^2+2D+2)}(1+3x+x^2)$$

$$y_p = \frac{1}{2\left(1+D+\frac{1}{2}D^2\right)}(1+3x+x^2)$$

~~$1+3x+$~~ ~~x^2+3x+1~~

~~x^2+3~~

$$y_p = \frac{1}{2}\left\{1-\left(D+\frac{D^2}{2}\right)+\left(D+\frac{D^2}{2}\right)^2\right\}(1+3x+x^2)$$

$$\frac{1}{2}\left\{1-D+\frac{D^2}{2}+D^2\right\}(1+3x+x^2)$$

$$= \frac{1}{2}\left\{1-D+\frac{D^2}{2}\right\}(1+3x+x^2)$$

$$= \frac{1}{2}\left\{1+3x+x^2-0-3-2x+0+0+1\right\}$$

$$= \frac{1}{2}\left\{x^2+x-1\right\}$$

$$\Rightarrow \frac{x^2+x-1}{2}.$$

$$y_p = \frac{x^2+x-1}{2}.$$

$$y = e^{-x}(C_1\cos x + C_2\sin x) + \frac{x^2+x-1}{2}.$$

$$\begin{array}{r|l} & \tfrac{1}{2}x^2+\tfrac{1}{2}x+\tfrac{1}{2} \\ \hline 2+2D+D^2 & x^2+3x+1 \\ & x^2+2x+1 \\ \hline & x+0 \\ & x+1 \\ \hline & =0 \end{array}$$

Scanned with OKEN Scanner

iii) $y''+y' = x^2+2x+4$

$\Rightarrow (D^2+D)y = x^2+2x+4$

To find y_c:

AE: $m^2+m=0$

$\Rightarrow m(m+1)=0$

$\Rightarrow m=0,-1$.

$y_c = c_1e^{0x}+c_2e^{-x} = c_1+c_2e^{-x}$.

To find y_p:

$y_p = \frac{1}{f(D)}\phi(x)$

$= \frac{1}{D^2+D}(x^2+2x+4)$

$= \frac{1}{D(1+D)}(x^2+2x+4)$

$= \frac{1}{D}\left(\frac{1}{1+D}\right)(x^2+2x+4)$

$= \frac{1}{D}\left[1-D+D^2\right](x^2+2x+4)$

$= \frac{1}{D}\left[x^2+2x+4-2x+2+2\right]$

$= \frac{1}{D}\left[x^2+4\right]$

$= \int(x^2+4)\,dx$

$y_p = \frac{x^3}{3}+4x$.

$y = c_1+c_2e^{-x}+\frac{x^3}{3}+4x$.

$$
\begin{array}{r|l}
 & x^2+4 \\
\hline
(1+D) & x^3+2x+4 \\
 & x^3+x^2 \\
\hline
 & -x^2+2x+4 \\
 & -x^2-x \\
\hline
 & 3x+4 \\
 & 3x+3 \\
\hline
 & 1
\end{array}
$$

$\frac{1}{D}\int(x^2+4)\,dx = \frac{x^3}{3}+4x$

y_p

Scanned with OKEN Scanner

PAGE NO.
DATE / /

N) $y'' - 4y' + 3y = 9x^2 + 4$

$\Rightarrow (D^2 - 4D + 3)y = 9x^2 + 4$

To find y_c:

A.E: $m^2 - 4m + 3$ $\quad +x-3$

$m = 1, 3$.

$y_c = C_1 e^x + C_2 e^{3x}$.

To find y_p:

$y_p = \frac{1}{f(D)} \phi(x)$

$y_p = \frac{1}{D^2 - 4D + 3} (9x^2 + 4)$

$= \frac{1}{3\left(1 - \frac{4}{3}D + \frac{D^2}{3}\right)} (9x^2 + 4)$

$= \frac{1}{3}\left(1 - \left(\frac{-4}{3}D + \frac{D^2}{3}\right) + \left(\frac{-4}{3}D + \frac{D^2}{3}\right)^2\right)(9x^2 + 4)$

$= \frac{1}{3}\left(1 + \frac{4}{3}D - \frac{D^2}{3} + \frac{16}{9}D^2\right)(9x^2 + 4)$

$= \frac{1}{3}\left(9x^2 + 4 + \frac{4}{3} \times \frac{3}{9}(18x) + \frac{13}{9} \times \frac{2}{18}\right)$

$= \frac{9x^2 + 4 + 24x + 26}{3}$

$= \frac{9x^2 + 24x + 30}{3} = 3x^2 + 8x + 10$.

$y = C_1 e^x + C_2 e^{3x} + 3x^2 + 8x + 10$.

Scanned with OKEN Scanner

v) $2y'' + 2y' + 3y = x^2 + 2x - 1$

$\rightarrow (2D^2 + 2D + 3)y = x^2 + 2x - 1$

To find y_c:

AE:

$2m^2 + 2m + 3 = 0$

$\Rightarrow m = \frac{-1}{2} \pm \frac{\sqrt{5}}{2} i$ $\quad \alpha = -1/2 \quad \beta = \sqrt{5}/2$

$y_c = e^{-1/2}\left(C_1 \cos\frac{\sqrt{5}}{2}x + C_2 \sin\frac{\sqrt{5}}{2}x\right)$

To find y_p: $\quad y_p = \frac{1}{f(D)} \phi(x)$

$y_p = \frac{1}{f(D)}(x^2 + 2x + 1)$

$= \frac{1}{3 + 2D + 2D^2}(x^2 + 2x - 1)$

$= \frac{1}{3\left(1 + \frac{2}{3}D + \frac{2}{3}D^2\right)}(x^2 + 2x + 1)$

t

$= \frac{1}{3}\left(1 - \frac{2}{3}D - \frac{2}{3}D^2 + \frac{4}{9}D^2\right)(x^2 + 2x + 1)$

$= \frac{1}{3}\left[1 - \frac{2}{3}D - \frac{2}{9}D^2\right](x^2 + 2x + 1)$

$= \frac{1}{3}\left(x^2 + 2x - 1 - \frac{2}{3}.2x - \frac{2}{3}.2 - \frac{2}{9}.2\right)$

$= \frac{1}{3}\left(x^2 + 2x - 1 - \frac{4}{3}x - \frac{4}{3} - \frac{4}{9}\right)$

$\frac{16}{9} + \frac{2}{9}$

$= \frac{1}{3}\left(x^2 + \frac{2x}{3} - \frac{25}{9}\right)$

$y_p = \frac{x^2}{3} + \frac{2x}{9} - \frac{25}{27}$

$y = y_c + y_p$

$y = e^{-1/2}\left(C_1 \cos\frac{\sqrt{5}}{2}x + C_2 \sin\frac{\sqrt{5}}{2}x\right) + \frac{x^2}{3} + \frac{2x}{9} - \frac{25}{27}$

Scanned with OKEN Scanner

PAGE NO.
DATE / /

Solve the foll. D.E.

1) $y'' + 4y = 2\sin^2 x$

$\rightarrow (D^2+4)y = 2\sin^2 x$

$\cos 2x = 1 - 2\sin^2 x$

$(D^2+4)y = 1 - \cos 2x$

To find y_c.

A.E $m^2+4=0$

$m = \pm 2i$ $\quad \alpha = 0, \beta = 2$

$y_c = c_1 \cos 2x + c_2 \sin 2x$

To find y_p: $y_p = \frac{1}{f(D)} \phi(x)$

$y_p = \frac{1}{f(D)} (1 - \cos 2x)$

$y_p = \frac{1}{f(D)}(1) - \frac{1}{f(D)} \cdot \cos 2x$

$y_p = \frac{1}{f(a)}(e^{0x}) - \frac{1}{f(a)}(\cos 2x)$

I) $\frac{1}{D^2+4}$ $\quad a = 0$

$f(a) = 4$

$\frac{1}{4}$.

II) $a = 2$ sub $D^2 = -4$

$f(D) = D^2 + 4$

$= -4 + 4 = 0$

$f'(D) = 2D$

$\frac{1}{2D} \cos 2x$

$= \frac{x}{2} \int \cos 2x \, dx$

$= \frac{x}{2} \frac{\sin 2x}{2} = \frac{x \sin 2x}{4}$.

$y = c_1 \cos 2x + c_2 \sin 2x + \frac{1}{4} + \frac{x \sin 2x}{4}$.

Scanned with OKEN Scanner

ii) $y'' + 3y' + 2y = 3e^{-2x} + \cos 3x$

$\rightarrow (D^2 + 3D + 2)y = 3e^{-2x} + \cos 3x$

To find y_c:

$m^2 + 3m + 2 = 0$

$m = -1, -2$.

$y_c = C_1 e^{-x} + C_2 e^{-2x}$

To find y_p:

$y_p = \frac{1}{f(D)} [3e^{-2x} + \cos 3x]$

$= \frac{1}{f(D)} 3e^{-2x} + \frac{1}{f(D)} \cos 3x$

I) $= \frac{1}{a}$ ✓ II) $a^2 = -9$

$a = -2$.

$\frac{1}{f(D)} \frac{1}{4 - 6 + 2}$

$f'(D) = 2D + 3$

$= -4 + 3 = -1$

$\frac{1}{D}(3e^{-2x})$

$= -3xe^{-2x}$

$f(D) = D^2 + 3D + 2 = -9 + 3D + 2$

$= -7 + 3D$.

$= \frac{1}{-7 + 3D} (\cos 3x)$

$\frac{3D + 7}{9D^2 - 49} \cos 3x$

$\frac{3D + 7}{-81 - 49} \cos 3x$

$= -\frac{(3D + 7)}{130} (\cos 3x)$

$= -\frac{-9\sin 3x + 7\cos 3x}{130}$

$= \frac{9\sin 3x - 7\cos 3x}{130}$.

Scanned with OKEN Scanner

iii] $(D^3-D)y = (2x+1) + 4\cos x$

→ To find y_c:

$m^3 - m = 0$

$m(m^2-1) = 0$

$\Rightarrow m = 0, 1, -1$

$y_c \Rightarrow c_1 + c_2 e^x + c_3 e^{-x}$

To find y_p:

$y_p = \frac{1}{f(D)} \phi(x)$

$= \frac{1}{f(D)} (2x+1) + \frac{1}{f(D)} (4\cos x)$

$= \frac{1}{D^3-D} (2x+1) + \frac{1}{D^3-D} (4\cos x)$

I II

I): $\frac{1}{D(D^2-1)} (2x+1) = \frac{1}{-D(1-D^2)} (2x+1)$

$= \frac{-1}{D} \left(1+D^2\right) (2x+1) = \frac{1}{D} (2x+1)$

$= -\int (2x+1)\,dx = -(x^2+x)$.

$= -x^2 - x$.

II): $\frac{1}{D(D^2-1)} 4\cos x = \frac{1}{D} (1+D^2)\, 4\cos x$

$= \frac{1}{D} (4\cos x + 4\cos x) =$

Ans

16/10

DATE / /

METHOD OF VARIATION OF PARAMETERS

- This is a general method to solve a 2nd order linear Differential Equation of the form $y'' + py' + qy = \phi(x)$ where p, q are functions of x.
Here particular integral is given by the formula,

$$y_p = -y_1 \int \frac{y_2 \phi(x)}{W} dx + y_2 \int \frac{y_1 \phi(x)}{W} dx$$

where y_1, y_2 are solutions of $f(D)y = 0$ (i.e. y_c)

$W = \begin{vmatrix} y_1 & y_2 \\ y_1' & y_2' \end{vmatrix}$ is called wronskian of y_1 & y_2.

Q) Solve the following D.E. by the method of variation of parameters.

i) $y'' + 4y = \tan 2x$

→ $(D^2 + 4)y = \tan 2x$.

Auxilliary Eqn: $m^2 + 4 = 0$

$\Rightarrow m = \pm 2i \Rightarrow \alpha = 0, \beta = 2$

$y_c = C_1 \cos 2x + C_2 \sin 2x$

To find y_p by variation of parameters,

$$y_p = -y_1 \int \frac{y_2 \phi(x)}{W} dx + y_2 \int \frac{y_1 \phi(x)}{W} dx$$

$\phi(x) = \tan 2x$; $y_1 = \cos 2x$; $y_2 = \sin 2x$

$y_1' = -2\sin 2x$; $y_2' = 2\cos 2x$

$$W = \begin{vmatrix} y_1 & y_2 \\ y_1' & y_2' \end{vmatrix} = \begin{vmatrix} \cos 2x & \sin 2x \\ -2\sin 2x & 2\cos 2x \end{vmatrix} = 2\cos^2 2x + 2\sin^2 2x = 2.$$

$\therefore \boxed{W = 2}$

Scanned with OKEN Scanner

$$y_p = -\cos 2x \int \frac{\sin 2x \cdot \tan(2x)}{2}\,dx + \sin 2x \int \frac{\cos 2x \cdot \tan 2x}{2}\,dx$$

$$y_p = -\frac{\cos 2x}{2}\int \frac{\sin^2 2x}{\cos 2x}\,dx + \frac{\sin 2x}{2}\int \sin 2x\,dx$$

$$y_p = -\frac{\cos 2x}{2}\int \frac{1-\cos^2 2x}{\cos 2x}\,dx + \frac{\sin 2x}{2}\left(\frac{-\cos 2x}{2}\right)$$

$$y_p = -\frac{\cos 2x}{2}\left[\int \sec 2x\,dx - \int \cos 2x\,dx\right] - \frac{\sin 2x\cos 2x}{4}$$

$$y_p = -\frac{\cos 2x}{2}\left[\frac{\log(\sec 2x+\tan 2x)}{2} - \frac{\sin 2x}{2}\right] - \frac{\sin 2x\cos 2x}{4}$$

$$y_p = -\frac{\cos 2x \log(\sec 2x+\tan 2x)}{4} + \frac{\sin 2x\cos 2x}{4} - \frac{\sin 2x\cos 2x}{4}$$

$$y_p = -\frac{\log(\sec 2x+\tan 2x)\cos 2x}{4} = -\frac{1}{4}\cos(2x)\log(\sec 2x+\tan 2x).$$

ii) $(D^2+1)y = \sec x \tan x$

→ To find y_c:

A.E. : $m^2+1=0$

$\Rightarrow m = \pm i$. $\quad \alpha = 0,\ \beta = 1$.

$y_c = C_1\cos x + C_2 \sin x$.

$\phi(x) = \sec x \tan x$; $y_1 = \cos x$; $y_2 = \sin x$

$y_1' = -\sin x$; $y_2' = \cos x$

$$W = \begin{vmatrix} y_1 & y_2 \\ y_1' & y_2' \end{vmatrix} = \begin{vmatrix} \cos x & \sin x \\ -\sin x & \cos x \end{vmatrix} = 1.$$

Scanned with OKEN Scanner

$$y_p = -y_1 \int \frac{y_2 \phi(x)}{W} dx + y_2 \int \frac{y_1 \phi(x)}{W} dx$$

$$y_p = \sin x \int \cos x \sec x \tan x \, dx - \cos x \int \sin x \sec x \tan x \, dx$$

$$y_p = \sin x \int \tan x \, dx - \cos x \int \tan^2 x \, dx$$

$$y_p = \sin x \log(\sec x) - \cos x \left(\int \sec^2 x \, dx - \int 1 \, dx \right)$$

$$y_p = \sin x \log(\sec x) - \cos x (\tan x - x)$$

$$y_p = \sin x \log(\sec x) - \sin x + x \cos x$$

$$y = y_c + y_p$$

$$y = C_1 \cos x + C_2 \sin x + \sin x \log(\sec x) - \sin x + x \cos x$$

iii) $(D^2 - 4D + 4) y = \frac{e^{2x}}{x}$

→ To find y_c:

A.E: $m^2 - 4m + 4 = 0$

$m = 2, 2$

$y_c = (C_1 + C_2 x) e^{2x} = C_1 e^{2x} + C_2 x e^{2x}$

To find y_p:

$$y_p = -y_1 \int \frac{y_2 \phi(x)}{W} dx + y_2 \int \frac{y_1 \phi(x)}{W} dx$$

$\phi(x) = e^{2x}/x$ $\quad y_1 = e^{2x}$ $\quad y_2 = x e^{2x}$

$y_1' = 2e^{2x}$ $\quad y_2' = 2x e^{2x} + e^{2x}$

$$W = \begin{vmatrix} y_1 & y_2 \\ y_1' & y_2' \end{vmatrix} = \begin{vmatrix} e^{2x} & x e^{2x} \\ 2e^{2x} & 2x e^{2x} + e^{2x} \end{vmatrix} = 2x e^{4x} + e^{4x} - 2x e^{4x}$$

$$= e^{4x}$$

Scanned with OKEN Scanner

$$y_p = -y_1 \int \frac{y_2 \phi(x)}{W} dx + y_2 \int \frac{y_1 \phi(x)}{W} dx$$

$$y_p = -e^{2x} \int \frac{x e^{2x} e^{2x}}{x e^{4x}} dx + x e^{2x} \int \frac{e^{2x} e^{2x}}{x e^{4x}} dx$$

$$y_p = -e^{2x} x + x e^{2x} \log(x)$$

$$y_p = -x e^{2x} + x e^{2x} \log(x)$$

$$y_p = x e^{2x} (\log x - 1)$$

$$y = y_c + y_p$$
$$= c_1 e^{2x} + c_2 x e^{2x} + x e^{2x} (\log x - 1)$$
$$= e^{2x} (c_1 + c_2 x + x(\log x - 1))$$

iv) $y'' + 4y' + 5y = e^{-2x}(\sec x)$

$\rightarrow (D^2 + 4D + 5) y = e^{-2x} \sec x$

To find y_c:

A.E: $m^2 + 4m + 5 = 0$

$m = -2 \pm i$ $\quad \alpha = -2, \beta = 1$ 1X5

$$y_c = e^{-2x}(c_1 \cos x + c_2 \sin x)$$

$$y_c = c_1 e^{-2x} \cos x + c_2 e^{-2x} \sin x$$

$\phi(x) = e^{-2x} \sec x$; $y_1 = e^{-2x} \cos x$; $y_2 = e^{-2x} \sin x$

$y_1' = -2e^{-2x} \cos x - e^{-2x} \sin x$; $y_2' = -2e^{-2x} \sin x + e^{-2x} \cos x$

$$W = \begin{vmatrix} y_1 & y_2 \\ y_1' & y_2' \end{vmatrix} = \begin{vmatrix} e^{-2x} \cos x & e^{-2x} \sin x \\ -2e^{-2x} \cos x - e^{-2x} \sin x & -2e^{-2x} \sin x + e^{-2x} \cos x \end{vmatrix}$$

Scanned with OKEN Scanner

$= -2e^{-4x}\sin x\cos x + e^{-4x}\cos^2 x + 2e^{-4x}\sin x\cos x + e^{-4x}\sin^2 x$

$= e^{-4x}$.

$\therefore y_p = -y_1\int \frac{y_2\,\phi(x)\,dx}{W} + y_2\int \frac{y_1\,\phi(x)\,dx}{W}$

$= -e^{-2x}\cos x\int \frac{e^{-2x}\sin x\, e^{-2x}\sec x}{e^{-4x}} + e^{-2x}\sin x\int \frac{e^{-2x}\cos x\, e^{-2x}\sec x}{e^{-4x}}$

$\therefore -e^{-2x}\cos x\log|\sec x| + e^{-2x}\sin x\, x$.

$y_p = e^{-2x}(\cos x\log|\sec x| + x\sin x)$.

$y = y_c + y_p$

$= e^{-2x}(c_1\cos x + c_2\sin x) + \cos x\log|\sec x| + x\sin x)$.